MIZORAM PUBLIC SERVICE COMMISSION

COMPETITIVE EXAMINATION FOR RECRUITMENT TO THE POST OF ASSISTANT DIRECTOR (FOOD & DRUGS) UNDER HEALTH & FAMILY WELFARE DEPARTMENT, GOVERNMENT OF MIZORAM. (NOVEMBER, 2025)

TECHNICAL PAPER-III

Time Allowed : 3 hours FM : 200

SECTION - A *(Conventional Type) (100 Marks)*

This Section should be answered only on the Answer Sheet provided.
Marks for each question is indicated against it.

1. Define enzyme immobilization. Elaborate various methods for enzyme immobilization and its application. **(2+8=10)**

2. Describe the general procedure for preparation of bacterial vaccines. Write a note on processing of Whole Human Blood. **(6+4=10)**

3. Discuss the types of documents maintained in a pharmaceutical industry with special reference to Master Formula Record, Batch Formula Record, and Standard Operating Procedures (SOPs). **(4+3+3=10)**

4. Explain the concept of technology transfer, as per WHO Guidelines. What are the principles and steps to be taken for NABL accreditation? **(5+5=10)**

5. Describe the design and layout of an aseptic area with appropriate labelled diagram. Explain the importance of laminar airflow equipment and discuss the various sources of contamination. **(4+3+3=10)**

6. Describe the general methods for the preparation of bacterial and viral vaccines, including toxoids and antitoxins. **5+5=10**

7. Define *Therapeutic Drug Monitoring (TDM).* Explain the need and importance of TDM in clinical practice. Discuss the factors to be considered during TDM considering the Indian scenario. **(2+4+4=10)**

8. Define patient counselling. Explain in detail the steps involved in effective patient counselling and describe the special situations where pharmacist's intervention and counselling are crucial. **(2+4+4=10)**

9. Write a note on preventive medicine and its importance in public health. Discuss the general principles of prevention and control of communicable and non-communicable diseases. **(4+6=10)**

10. Write the WHO guidelines of documentation commonly required for technology transfer of Pharmaceutical Manufacturing. **(10)**

SECTION - B *(Multiple Choice questions) (100 Marks)*

All questions carry equal mark of 2 each. Attempt all questions.
*This Section should be answered only on the **OMR Response Sheet** provided.*

1. Which element is essential for implementing successful standard operating procedures?
(a) Making SOPs accessible only to management
(b) Elimination of all previous protocol immediately
(c) Regular training for staff
(d) Frequent revisions of SOPs

2. Master formula record is prepared and endorsed by the competent technical staff like manufacturing chemist and ____chemist.
(a) Quality control (b) Analytical
(c) Production (d) Formulation

3. Quality audit may include questionnaires on Good Manufacturing process (GMP) requirements covering
(a) Personnel (b) Documentation
(c) Recall procedures (d) Finished products

4. Regulatory audit is also known as _____.
(a) First party audit (b) Second party audit
(c) Third party audit (d) Fourth party audit

5. In technology transfer, the Sending Unit (SU) should provide:
(a) Its support function
(b) The necessary validation documentation for the process
(c) Criteria and information on hazards and critical steps associated with the products, process or method to be transferred.
(d) To serves as a basis for Quality Risk Management (QRM) exercise at Receiving Unit (RU)

6. Which of the following is a phase of technology transfer process considered as the last phase?
(a) Development Phase (b) Production Phase
(c) Research Phase (d) Quality testing Phase

7. Which Government has authorized NABL as the sole accreditation body for testing and calibrations Laboratory?
(a) US (b) Japan
(c) India (d) All of the above

8. Which standard is commonly used as a basis for NABL accreditation of testing and calibration laboratories?
(a) ISO/IEC 17025 (b) ISO 9001
(c) ISO 14001 (d) ISO 45001

9. Which of the following clean rooms grade is used for aseptic filling operation?
(a) GRADE A (b) GRADE B
(c) GRADE C (d) GRADE D

10. Which of the following is the most accurate method for microbial assay of antibiotics?
(a) Physical assay (b) Chemical assay
(c) Biological assay (d) Chemical and biological assay

11. Which of the following is true for a transducer in biosensor?

(a) It displays the signal

(b) It converts the interaction between the analytes and the immobilized enzymes into a measurable signal

(c) It is used as bioreceptors

(d) It is an optional part which is rarely used

12. Which of the following does not produce 'Proteases'?

(a) Bacillus (b) Rhizopus

(c) Mucor (d) Bacillus coagulans

13. Which type of immunoglobulin provides local immunity?

(a) IgA (b) IgG

(c) IgD (d) IgM

14. Once blood is collected, the constituents should be separated within:

(a) 3-5 hours (b) 5-8 hours

(c) 7-9 hours (d) 10-13 hours

15. The shelf life of dried human plasma is:

(a) 1 year (b) 2 years

(c) 5 years (d) 10 years

16. Which of the following vaccine is an example of live attenuated bacterial vaccine?

(a) Cholera (b) Anthrax

(c) Plague (d) Typhoid

17. The amount of a drug product that should be kept in the pharmacy shelf is called:

(a) Standing Order (b) Basic Stock

(c) PAR or Maximum stock level (d) Restricted distribution

18. A form that tracks the medications administered to a patient is called:

(a) Open Formulary (b) Patient wrist band

(c) Closed Formulary (d) Medical administration record

19. The basic principle of therapeutic drug monitoring is:

(a) To measure patient's drug blood concentration

(b) To measure patient urine drug concentration

(c) Limited role in toxicology

(d) To study adverse effect of drugs

20. The therapeutic index is defined as:

(a) The concentration of drug in plasma which optimizes therapy

(b) The amount of reduction of pathologic symptoms obtained by use of a specific drug

(c) The ranking of the effect of the drug versus other drugs for the same disease

(d) The ratio between the average toxic dose and the average therapeutic dose

21. In India, therapeutic drug monitoring was introduced in the year:

(a) 1980 (b) 1985

(c) 1986 (d) 1990

22. What is the primary method recommended by World Health Organization to prevent dengue infection?

(a) Taking an over-the-counter pain reliever when symptoms appear
(b) Drinking lots of fluids
(c) Protecting yourself from mosquito bites
(d) Avoiding travel to endemic areas

23. The physiological significance of blood glucose monitoring in diabetic patient is:

(a) Confirm medication adherence
(b) Monitor the efficacy of hypoglycaemic agent
(c) Track potential ADRs of Antibiotics
(d) Ensure liver function is not compromised

24. Which of these is the goal of cancer screening?

(a) To treat cancer once it is diagnosed
(b) To detect cancer at an early stage
(c) To prevent the development of cancer cells
(d) To manage the side effects of chemotherapy

25. _____ is the addition of a known amount of a compound to a standard, sample or placebo, typically for the purpose of confirming the performance of an analytical procedure.

(a) Validation
(b) Enriching
(c) Leaching
(d) Spiking

26. Consider these statements on the generic drug approval process under USFDA:

I. Clinical trials are not mandatory.

II. They may submit a marketing authorization application to the FDA in the form of an Abbreviated New Drug Application (ANDA).

(a) Only I is true
(b) Only II is true
(c) Both I and II are false
(d) Both I and II are true

27. Certificate of Pharmaceutical Products (CoPP) is issued by _____.

(a) The regulatory authority of the exporting country to the importing country
(b) The regulatory authority of the importing country to the exporting country
(c) WHO only
(d) USFDA only

28. Within the classical drug development, compliance with GLP is required in _____.

(a) Stage 1 (Discovery stage)
(b) Stage 2 (Non-clinical stage)
(c) Stage 3 (Clinical stage)
(d) Stage 4 (Post-approval stage)

29. This individual will have to assume full responsibility for the GLP compliance of all activities within the study and is the single most important individual in GLP study.

(a) Principal Investigator
(b) Research Scientist
(c) Volunteer
(d) Study Director

30. According to WHO GLP guidelines, '*dirty corridors*' in animal facilities means _____.

(a) A corridor for the movement of used supplies
(b) A corridor for the movement of new supplies
(c) A corridor for the movement of dead animals only
(d) A corridor for the movement of laboratory staff

31. Restriction enzymes are called as

(a) Molecular scissor
(b) Molecular glue
(c) Ligase enzyme
(d) Polymerase enzyme

32. For a pharmaceutical product on the market, the generally accepted retention period of the records containing all original observations, including calculations and derived data, calibration, validation and verification etc. is _____.

(a) Shelf-life plus 25 years
(b) Shelf-life plus 50 years
(c) Shelf-life plus 10 years
(d) Shelf-life plus 1 year

33. HEPA filters remove particles as small as:

(a) 0.1 µm
(b) 0.3 µm
(c) 1.0 µm
(d) 5.0 µm

34. Enzyme immobilization increases:

(a) Enzyme solubility
(b) Enzyme activity in solution
(c) Enzyme reusability and stability
(d) Reaction rate always

35. A glucose biosensor uses which enzyme?

(a) Urease
(b) Glucose oxidase
(c) Peroxidase
(d) Catalase

36. Vaccines should generally be stored at:

(a) 0°C
(b) 2–8°C
(c) 10–15°C
(d) Room temperature

37. A commonly used preservative in vaccines is:

(a) Benzalkonium chloride
(b) Alcohol
(c) Formaldehyde
(d) Thiomersal

38. The most common microbial source for industrial production of amylase is:

(a) *Escherichia coli*
(b) *Aspergillus oryzae*
(c) *Penicillium chrysogenum*
(d) *Saccharomyces cerevisiae*

39. During the preparation of bacterial vaccines, the inactivation of bacteria is commonly achieved using:

(a) Phenol or formaldehyde at controlled temperature
(b) UV radiation alone
(c) High-pressure homogenization
(d) Freezing and thawing

40. Ideal characteristics of a plasma substitute include all *except*:

(a) Non-toxic and non-pyrogenic
(b) Osmotically active
(c) Readily metabolized to water and CO_2
(d) Antigenic with long half-life

41. Lipase production is enhanced when the medium contains:

(a) Carbohydrates only
(b) Inducers like olive oil or triglycerides
(c) Ammonium salts only
(d) Alcohols

42. The unit dose drug distribution system primarily aims to:

(a) Reduce nursing workload
(b) Minimize medication errors and wastage
(c) Store large quantities of drugs in wards
(d) Replace pharmacy staff

43. Which of the following is considered a controlled drug?

(a) Paracetamol (b) Morphine

(c) Amoxicillin (d) Metformin

44. Which of the following drugs commonly require TDM?

(a) Digoxin, Vancomycin, Phenytoin (b) Paracetamol, Ibuprofen, Aspirin

(c) Omeprazole, Pantoprazole (d) Vitamin D, Calcium

45. The most common cause of medication non-adherence is:

(a) Poor drug stability

(b) Complex drug regimens and side effects

(c) Pharmacy location

(d) Nurse unavailability

46. Pharmacists can improve medication adherence by:

(a) Reducing consultation time

(b) Changing drug brand only

(c) Patient education, regimen simplification, reminder systems

(d) Avoiding follow-ups

47. A major challenge in treating drug addiction pharmacologically is:

(a) Enhancing drug metabolism

(b) Preventing relapse due to craving triggered by environmental cues

(c) Administering vaccines

(d) Improving kidney clearance

48. Which population is most at risk of non-adherence due to cognitive impairment?

(a) Young adults (b) Geriatric patients

(c) Healthy children (d) Healthcare professionals

49. Directly Observed Therapy (DOT**)** is an effective strategy for improving adherence in:

(a) Hypertension (b) Congestive Heart Failure

(c) Tuberculosis (d) Vitamin supplementation

50. Good Laboratory practice comes under which schedule?

(a) L1 (b) L2

(c) L3 (d) L4

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